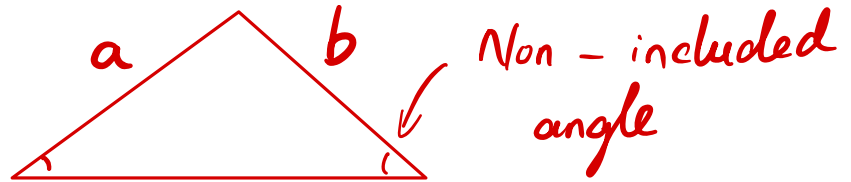


21/06/2026

Sine Rule



The **sine rule** is used to find unknown side lengths or angles of a triangle in the following two situations:

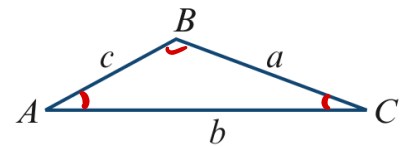
- 1 one side and two angles are given ← Case 1
- 2 two sides and a non-included angle are given (that is, the given angle is not 'between' the two given sides). ← Case 2

In the first case, the triangle is uniquely defined up to congruence. In the second case, there may be two triangles.

Labelling triangles

The following convention is used in the remainder of this chapter:

- Interior angles are denoted by uppercase letters.
- The length of the side opposite an angle is denoted by the corresponding lowercase letter.

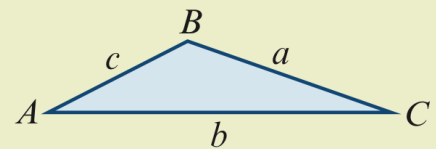


For example, the magnitude of angle BAC is denoted by A , and the length of side BC is denoted by a .

Sine rule

For triangle ABC :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

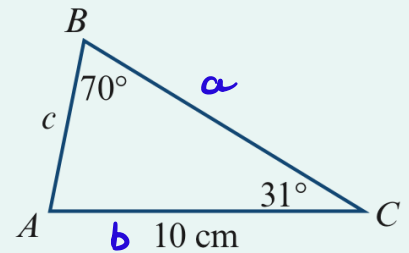


Case 1

► One side and two angles

Example 3

Use the sine rule to find the length of AB .



$$\frac{a}{\sin A} = \frac{b \checkmark}{\sin B \checkmark} = \frac{c ?}{\sin C \checkmark}$$

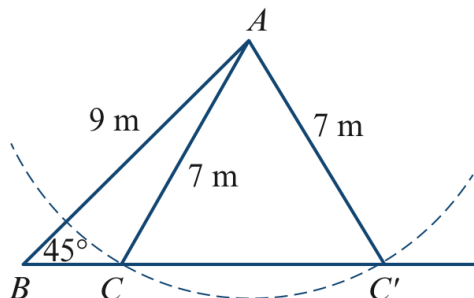
$$\frac{10}{\sin(70^\circ)} = \frac{c}{\sin(31^\circ)}$$

$$c = \frac{10 \sin(31^\circ)}{\sin 70^\circ} = 5.48 \text{ cm}$$

Case 2

► Two sides and a non-included angle

Suppose that we are given the two side lengths 7 m and 9 m and a non-included angle of 45° . There are two triangles that satisfy these conditions, as shown in the diagram.

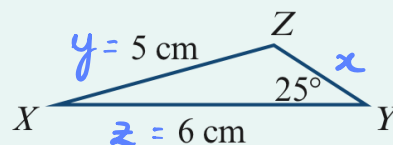


Warning

- When you are given two sides and a non-included angle, you must consider the possibility that there are two such triangles.
- An angle found using the sine rule is possible if the sum of the given angle and the found angle is less than 180° .

Example 4

Use the sine rule to find the magnitude of angle XZY in the triangle, given that $Y = 25^\circ$, $y = 5$ cm and $z = 6$ cm.



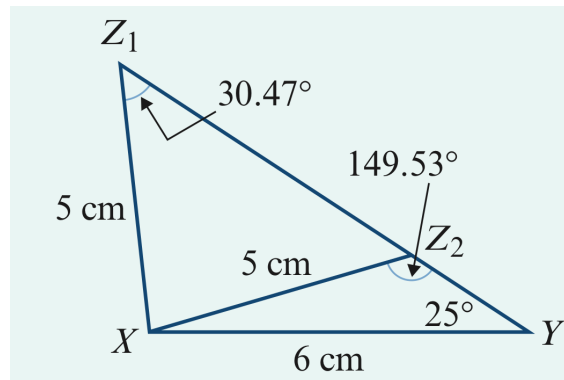
$$\frac{x}{\sin X} = \frac{y}{\sin Y} = \frac{z}{\sin Z}$$

$$\frac{5}{\sin(25^\circ)} = \frac{6}{\sin Z}$$

$$\sin Z = \frac{6 \sin(25^\circ)}{5}$$

$$Z_1 = \sin^{-1}\left(\frac{6 \sin 25^\circ}{5}\right) = 30.47^\circ$$

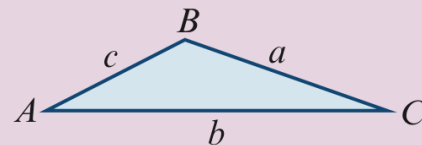
$$Z_2 = 180^\circ - Z_1 = 149.53^\circ$$



Section summary

■ **Sine rule** For triangle ABC :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

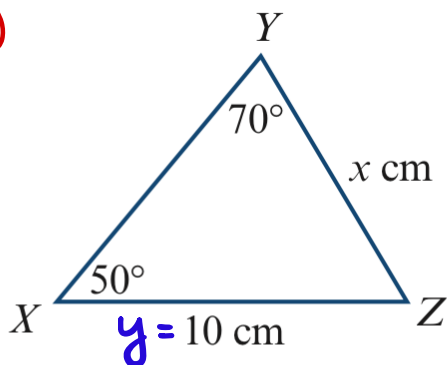


■ When to use the sine rule:

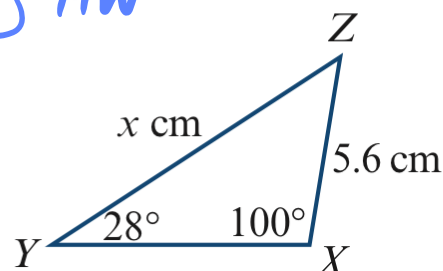
- one side and two angles are given (AAS) ← Case 1
- two sides and a non-included angle are given. ← Case 2

1 Find the value of the pronumeral for each of the following triangles:

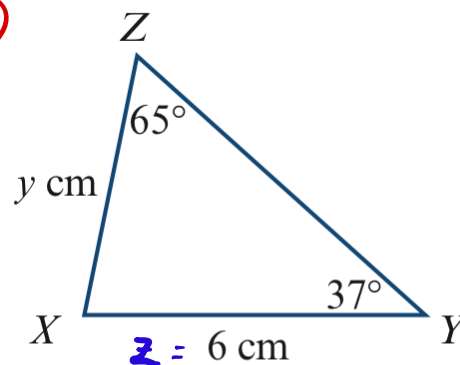
(a)



(c)

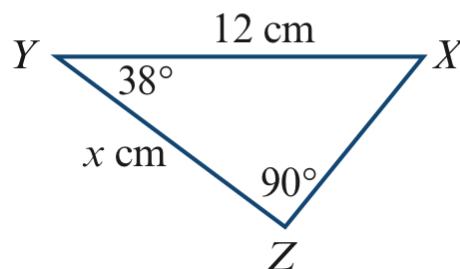


(b)



HW

(d)

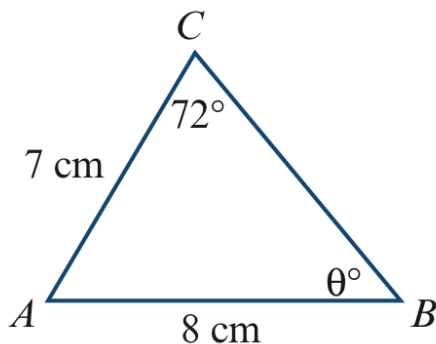


$$\begin{aligned} (a) \quad \frac{x}{\sin X} &= \frac{y}{\sin Y} \\ \frac{x}{\sin 50^\circ} &= \frac{10}{\sin 70^\circ} \\ x &= \frac{10 \sin 50^\circ}{\sin 70^\circ} \\ &= 8.15 \end{aligned}$$

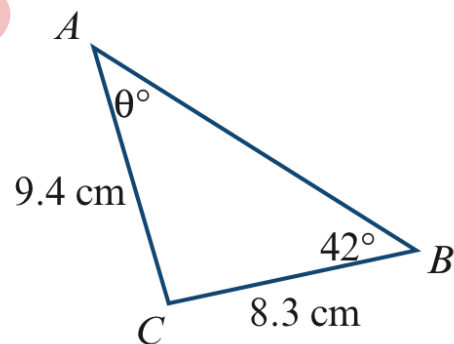
$$\begin{aligned} (b) \quad \frac{z}{\sin(Z)} &= \frac{y}{\sin(Y)} \\ \frac{6}{\sin 65^\circ} &= \frac{y}{\sin 37^\circ} \\ y &= \frac{6 \sin 37^\circ}{\sin 65^\circ} \\ &= 3.98 \end{aligned}$$

2 Find the value of θ for each of the following triangles:

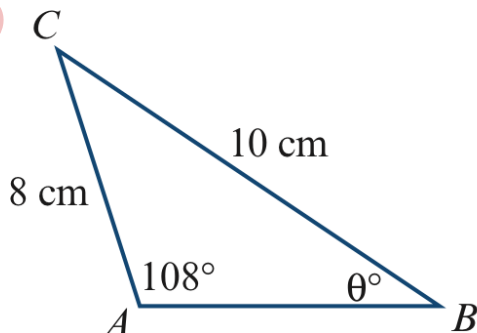
a



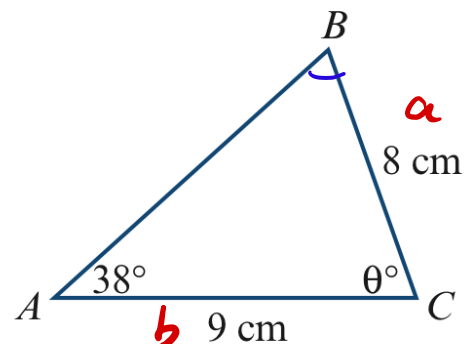
b



c



d



$$(d) \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{8}{\sin 38^\circ} = \frac{9}{\sin B}$$

$$\sin B = \frac{9 \sin 38^\circ}{8} = 0.6926$$

$$B_1 = 43.83^\circ$$

$$B_2 = 180^\circ - 43.83^\circ = 136.16^\circ$$

$$C_1 = 180^\circ - A - B_1 = 180^\circ - 38^\circ - 43.83^\circ = 98.17^\circ$$

$$C_2 = 180^\circ - A - B_2 = 180^\circ - 38^\circ - 136.16^\circ = 5.84^\circ$$

The cosine rule

The **cosine rule** is used to find unknown side lengths or angles of a triangle in the following two situations:

- 1 two sides and the included angle are given
- 2 three sides are given.

In each case, the triangle is uniquely defined up to congruence.

Cosine rule

$$b^2 = a^2 + c^2 - 2ac \cos B$$

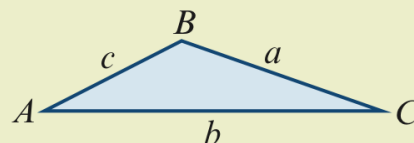
For triangle ABC :

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

or equivalently

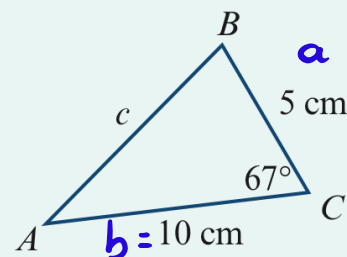
$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



► Two sides and the included angle

Example 5

For triangle ABC , find the length of AB in centimetres correct to two decimal places.

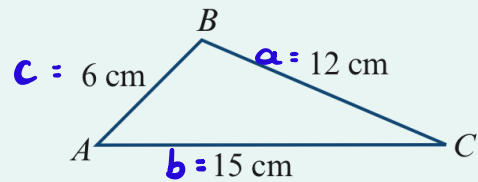


$$\begin{aligned} c^2 &= a^2 + b^2 - 2ab \cos C \\ &= 5^2 + 10^2 - 2 \times 5 \times 10 \cos(67^\circ) \\ &= 85.9268 \dots \\ c &= 9.27 \text{ cm} \end{aligned}$$

► Three sides

Example 6

Find the magnitude of angle ABC .



$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$b^2 = a^2 + c^2 - 2ac \cos B \quad \leftarrow$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$2ac \cos B = a^2 + c^2 - b^2$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{12^2 + 6^2 - 15^2}{2 \times 12 \times 6} = -0.3125$$

$$B = 108.21^\circ$$

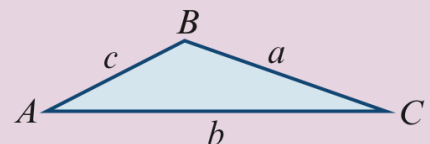
Section summary

■ **Cosine rule** For triangle ABC :

$$c^2 = a^2 + b^2 - 2ab \cos C \quad \text{or} \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

■ When to use the cosine rule:

- two sides and the included angle are given (SAS)
- three sides are given (SSS).



$$a^2 = b^2 + c^2 - 2bc \cos A$$

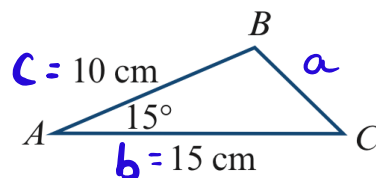
$$b^2 = a^2 + c^2 - 2ac \cos B$$

- 1 Find the length of BC .

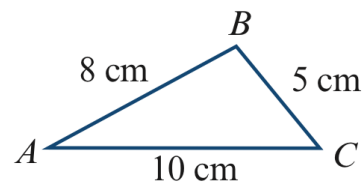
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$= 15^2 + 10^2 - 2 \times 15 \times 10 \cos(15^\circ)$$

$$a = 5.93 \text{ cm}$$



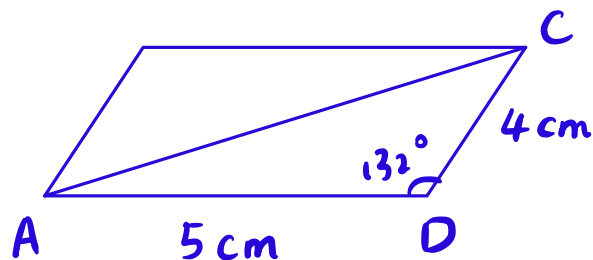
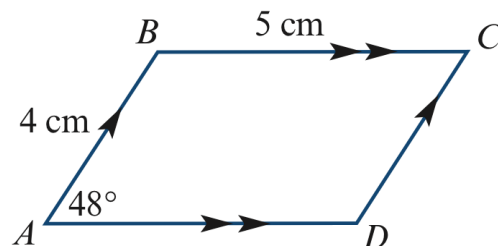
- 2 Find the magnitudes of angles ABC and ACB .



- 7 $ABCD$ is a parallelogram. Find the lengths of the diagonals:

a AC

b BD

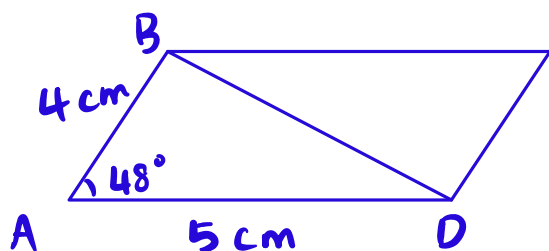


$$d^2 = a^2 + c^2 - 2ac \cos D$$

$$= 4^2 + 5^2 - 2 \times 4 \times 5 \cos 132^\circ$$

$$= 67.77$$

$$d = 8.23 \text{ cm} = AC$$



$$a^2 = b^2 + d^2 - 2bd \cos A$$

$$= 5^2 + 4^2 - 2 \times 5 \times 4 \cos 48^\circ$$

$$= 14.23$$

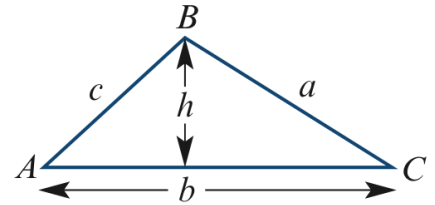
$$a = 3.77 \text{ cm}$$

$$BD = 3.77 \text{ cm}$$

The area of a triangle

The area of a triangle is given by

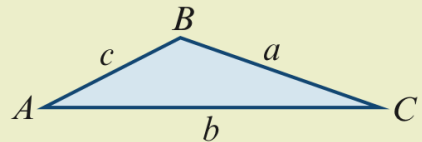
$$\begin{aligned}\text{Area} &= \frac{1}{2} \times \text{base length} \times \text{height} \\ &= \frac{1}{2}bh\end{aligned}$$



By observing that $h = c \sin A$, we obtain the following useful formula.

For triangle ABC :

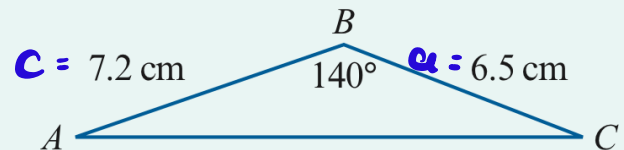
$$\text{Area} = \frac{1}{2}bc \sin A = \frac{1}{2}ac \sin B = \frac{1}{2}ab \sin C$$



That is, the area is half the product of the lengths of two sides and the sine of the angle included between them.

Example 7

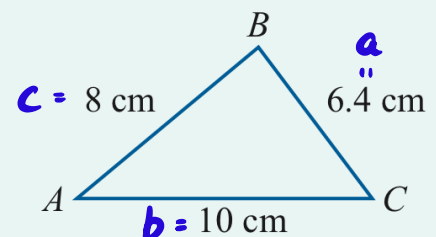
Find the area of triangle ABC shown in the diagram.



$$A = \frac{1}{2}ac \sin B = \frac{1}{2} \times 6.5 \times 7.2 \times \sin 140^\circ = 15.04 \text{ cm}^2$$

Example 8

Find the area of the triangle, correct to three decimal places.



$$c^2 = a^2 + b^2 - 2ab \cos C$$

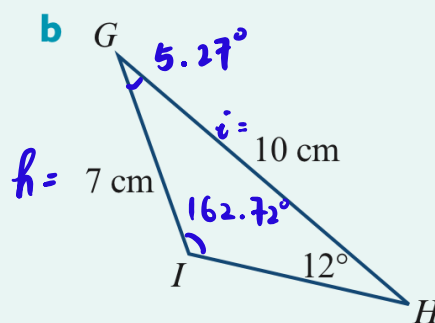
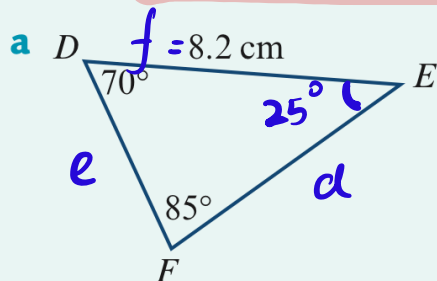
$$8^2 = 6.4^2 + 10^2 - 2 \times 6.4 \times 10 \cos C$$

$$\Rightarrow C = 53.04^\circ$$

$$A = \frac{1}{2}ab \sin C = \frac{1}{2} \times 6.4 \times 10 \sin 53.04^\circ = 25.57 \text{ cm}^2$$

Example 9

Find the area of each of the following triangles, correct to three decimal places:



$$(a) \quad \frac{\sin 85^\circ}{8.2} = \frac{\sin 25^\circ}{DF}$$

$$e = DF = 3.48$$

$$A = \frac{1}{2} ef \sin D$$

$$= \frac{1}{2} \times 3.48 \times 8.2 \sin 70^\circ$$

$$= 13.403 \text{ cm}^2$$

$$(b) \quad \frac{\sin I}{10} = \frac{\sin 12^\circ}{7}$$

$$\sin I = 0.2970$$

$$I = 162.72^\circ$$

theoretically I has two values, but from the diagram I is an obtuse angle, only need one.

$$G = 180^\circ - 12^\circ - 162.72^\circ$$

$$= 5.27^\circ$$

$$A = \frac{1}{2} hi \sin G$$

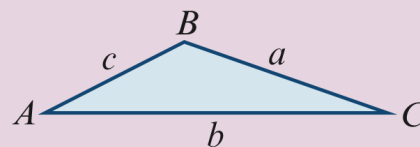
$$= \frac{1}{2} \times 7 \times 10 \sin 5.27^\circ$$

$$= 3.22 \text{ cm}^2$$

Section summary

For triangle ABC :

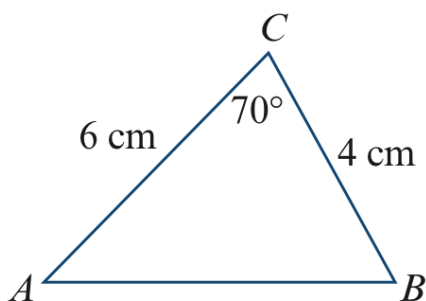
$$\text{Area} = \frac{1}{2}bc \sin A$$



That is, the area is half the product of the lengths of two sides and the sine of the angle included between them.

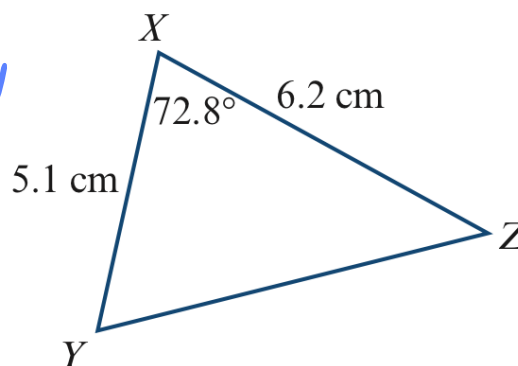
1 Find the area of each of the following triangles:

a



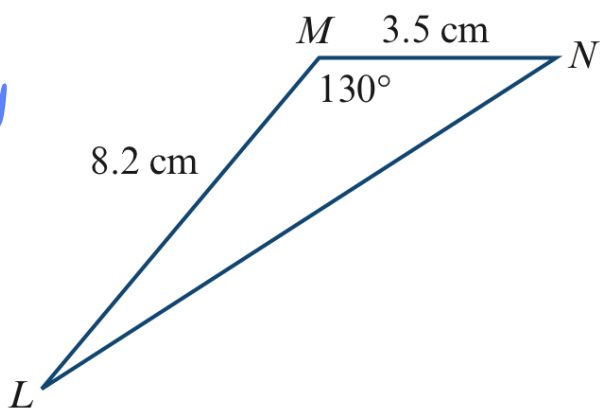
b

HW

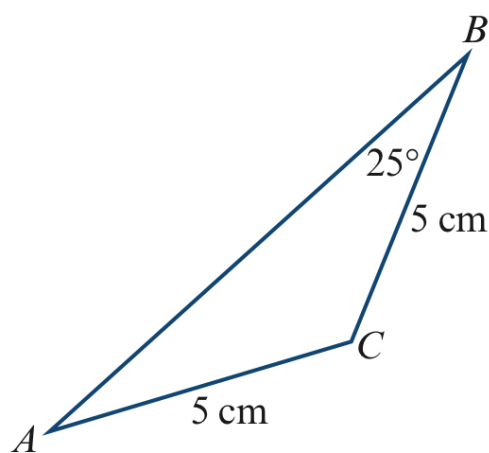


c

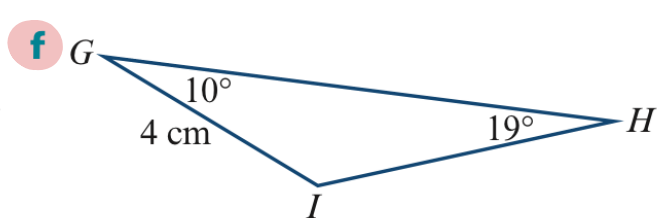
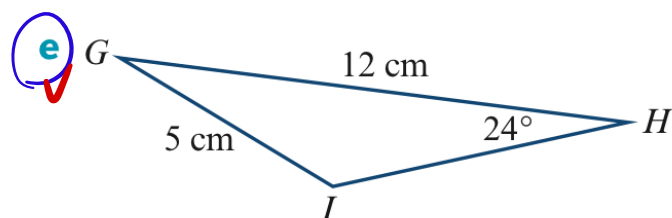
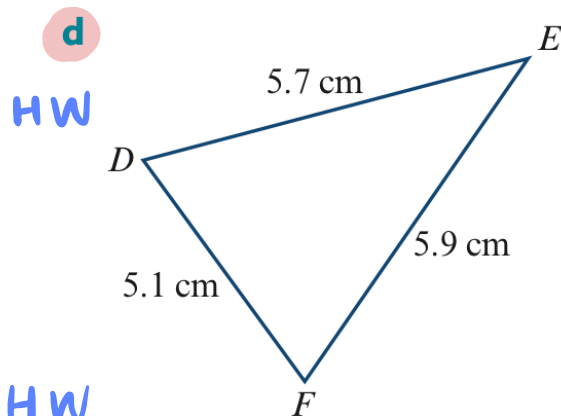
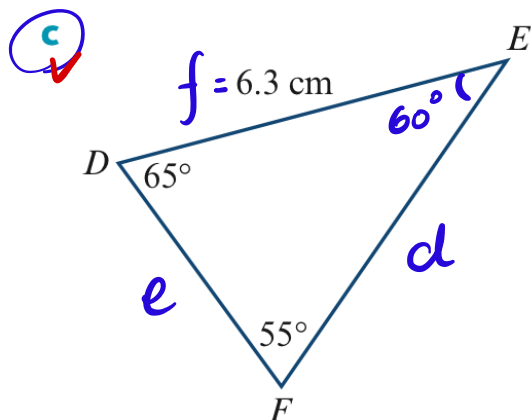
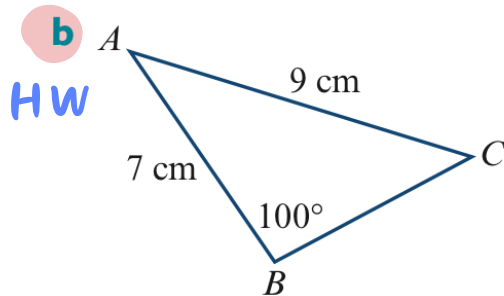
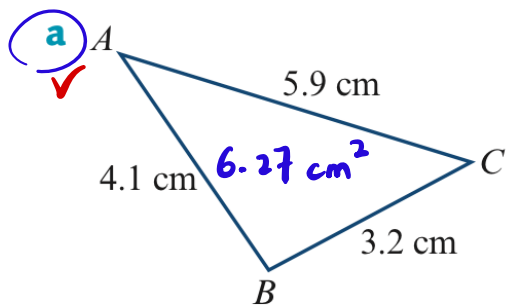
HW



d



2 Find the area of each of the following triangles, correct to three decimal places:



(c)
$$\frac{\sin 65^\circ}{EF} = \frac{\sin 55^\circ}{6.3}$$

$$d = EF = 6.97 \text{ cm}$$

$$A = \frac{1}{2} df \sin E = \frac{1}{2} \times 6.97 \times 6.3 \times \sin 60^\circ = 19.01 \text{ cm}^2$$